

Jude Vamenta

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EDUCATION

Tufts University, School of Engineering, Medford, MA

BS in Mechanical Engineering, Minor in Music Engineering

Expected May 2027

GPA: 3.97, Dean's List All Semesters

Relevant Coursework: Electronics and Controls, Materials and Manufacturing, Thermodynamics, Dynamics

EXPERIENCE

Trane Technologies, Commercial HVAC, Technical Sales Engineer Intern – Wilmington, MA May 2025 – August 2025

- Conducted 20+ HVAC system site visits (chillers, rooftop units, AHUs) to support equipment selection, system design, and applied thermodynamics analysis.
- Supported multi-million-dollar project proposals through cost estimation, performance modeling, and equipment data analysis (100+ assets).
- Implemented a Salesforce-based workflow for service agreements and renewals, improving operational efficiency and response time across the New England region.

Tufts University, Bray Lab Machine Shop, Shop Assistant – Medford, MA September 2025 – Present

- Trained on and operated lathes, mills, waterjet cutters, laser cutters, and 3D printers for fabrication and prototyping.
- Built proficiency in machining, CAD/CAM workflows, and shop safety protocols to support mechanical engineering projects.
- Assisted peers with hands-on manufacturing methods, reinforcing design-to-build execution.

Tufts Wilderness Orientation, Trip Leader – Medford, MA May 2025 – August 2025

- Led 7 first-year students on a 5-day backpacking trip, planning routes, meals, water, breaks, and managing group dynamics for a safe first-time experience.
- Trained students in wilderness skills and teamwork, teaching stove assembly, tarp setup, bear-bagging, and fostering communication while applying Wilderness First Aid and CPR/AED for safety.

Tufts University, Film Photography Laboratory, Darkroom Monitor – Medford, MA September 2024 – May 2025

- Provided technical support and troubleshooting for lab equipment, ensuring consistent operation and workflow for student projects.

PROJECTS

Tufts University Engineers Without Borders Greenhouse, Technical Lead sites.tufts.edu/ewb/greenhouse-project

- Led a team of 10 to design and build an Arduino-controlled autonomous watering system, integrating solenoid valves and moisture sensors.
- Developed weekly lesson plans to teach members CAD, C++, and prototyping workflows, meeting project deadlines within a \$400 budget.

Mystic River Watershed Association - Stormwater Infiltration Trench Model mysticriver.org/trenches

- Designed and fabricated a 3D-printed and laser-cut interactive model of a stormwater infiltration trench in CAD.
- Integrated custom parts to create a watertight, educational prototype demonstrating real-world hydrological concepts.

SKILLS

CAD: SOLIDWORKS, Fusion 360, OnShape

Fabrication/Prototyping: 3D Printing, Laser Cutting, Drill Press, Metal Lathe, Sand Casting, Water Jet, Sheet Metal

Programming/Controls: C++, MATLAB, Arduino/Microcontrollers